

Arnav Sinha

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Education

Vellore Institute of Technology

B.Tech., Computer Science and Engineering (AI/ML) | CGPA: 8.56/10.0

Vellore, India

Expected May 2027

Relevant Coursework: Data Structures & Algorithms, Software Engineering, Operating Systems, Computer Networks, DBMS, Machine Learning, Deep Learning, Reinforcement Learning, Machine Vision, Speech & Language Processing, Explainable AI

Experience

Inadev Corporation

Data Science Intern

May 2026 – Aug 2026

Kolkata, India (Hybrid)

SOW AI Extraction Service – Python, FastAPI, Docker, AWS Lambda, AWS Bedrock, Tesseract OCR

- Shipped a stateless REST microservice as a container image on AWS Lambda behind a load balancer, converting contract PDFs into a total value and dated billing schedule for the portal's revenue-tracking module; extraction failures return HTTP 200 with a FAILED status the caller can persist
- Confined the LLM to fact extraction and computed every total, date and compounding step in Python, keeping the figures reproducible; matched a 3-year 5%-escalating contract's own schedule to the rupee (INR 3.02 Cr, 36 cycles)
- Root-caused silent revenue loss where routing pages by character count let a scanned pricing page skip OCR; fixed it by OCR-ing every page, and returned nulls with review reasons on unquantifiable contracts rather than fabricated figures

Legal Contract RAG Assistant – LangGraph, MongoDB Atlas Vector Search, AWS Bedrock, LangSmith, Streamlit

- Built a citation-backed RAG system over a 48-contract legal corpus on MongoDB Atlas Vector Search with Gemini embeddings and a BGE cross-encoder reranker, tuned by a 189-question study with bootstrap 95% CIs (recall@5 0.815 against BM25 0.487)
- Replaced a single-shot chain with a LangGraph tool-calling agent whose two tools came from failure analysis of 35 measured retrieval failures, recovering 23/35 (66%) and enabling multi-document comparison
- Caught a 26% regression (42 of 162 questions) with a regression suite graded by an independent local model rather than self-scored, and halved it with a four-stage name-resolution ladder; traced every turn in LangSmith and shipped a provider-agnostic Streamlit chat UI over Ollama, Bedrock and Gemini

Projects

Ripple-Aware Satellite Collision Avoidance – Capstone Research

JAX, Multi-Agent RL (MAPPO), GNN, SGP4

- Measured from 620K real Space-Track orbit records (1,800 Starlink satellites, 11.9K manoeuvres found by a drag-aware detector) that every manoeuvre leaves a satellite's public catalogue entry $2.2\times$ less accurate, causing 120 false and 111 missed conjunction alerts against 136 genuine
- Built a GPU-accelerated JAX orbital simulator (two-body + J2 gravity, 837 real objects, 14.4M object-steps/s on a laptop GPU) showing that announcing burns at planning time removes 92% of induced false alarms
- Trained a 103K-parameter GNN policy with MAPPO and a Lagrangian safety constraint that, given fresh orbit data, left 11% fewer close approaches unresolved at equal fuel (paired $t = 4.67, 8/8$ held-out worlds); debugged critic-dominated gradients and entropy collapse over ~ 180 runs

GymPose-Lite – Distilled Real-Time Pose Estimation

PyTorch, Knowledge Distillation, OpenCV, GNN, Flask

- Distilled a 59M-parameter KeypointRCNN-ResNet50 teacher into a 9M-parameter MobileNetV3 student (6.5x smaller) on 148K teacher-labelled COCO person crops, after diagnosing that direct MSE training on sparse heatmaps stalled for 80+ epochs
- Added a graph-network refinement head over the 17-joint skeleton: 20.9px median joint error vs the teacher on 14.8K held-out crops, at ~ 11 ms/frame on a laptop GPU (~ 30 FPS on CPU)
- Built end-to-end gym form feedback: a state-machine rep counter with hysteresis jitter suppression, rule-based per-rep form scoring, and a Flask video-upload app

FinVoice – AI-Powered Financial Call Analytics

WhisperX, FinBERT, Presidio, Next.js

- Built a 5-stage ML pipeline running 10+ models on a 6 GB-GPU laptop (WhisperX ASR, Emotion2Vec, FinBERT sentiment, Qwen extraction), plus zero-shot scam classification (mDeBERTa) and an eval harness that surfaced silent failures
- Developed PII detection using Presidio with custom Indian recognizers (Aadhaar with Verhoeff checksum, PAN, UPI, IFSC) and a compliance engine covering RBI Fair Practice Code, KYC and DPDP Act 2023 in English, Hindi and Tamil

Publications & Achievements

- R. Bansal, S. Madan, **A. Sinha**, Karthik G M, "Ripple-Aware Cooperative Collision Avoidance for Satellite Mega-Constellations," accepted at *ICon INDIA 2026* (IEEE technically co-sponsored), Pondicherry University
- First Place, IEEE CS VIT Hackbattle 2025** (600+ participants): built TheBlinkingWorlds, a 2D platformer in Godot where real-time OpenCV blink detection switches the game world

Technical Skills

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL, HTML/CSS

AI/ML: PyTorch, JAX, TensorFlow, Keras, scikit-learn, HuggingFace Transformers, OpenCV, spaCy, NLTK, NumPy, Pandas, Reinforcement Learning (PPO, MAPPO), Knowledge Distillation

Generative AI: RAG, LangGraph, LangChain, LangSmith, MCP (Model Context Protocol), Vector Search, Embeddings, Cross-Encoder Reranking, LLM-as-Judge, RAGAS, AWS Bedrock, Ollama

Backend, Cloud & Data: FastAPI, Flask, REST APIs, Docker, Linux, Git/GitHub, AWS (Lambda, EC2, S3, Bedrock), React.js, Next.js, Node.js, Streamlit, MongoDB (Atlas Vector Search), PostgreSQL, MySQL, ChromaDB